Integrated Waste Management Knowledge Graph Architecture for German Academic Medical Centers

1. Introduction: The Imperative for Semantic Waste Management

The effective management of waste streams within German academic and medical institutions is not merely a logistical challenge but a complex legal, environmental, and safety mandate. Universities act as unique generators of highly diverse waste types—ranging from standard municipal waste to high-risk infectious agents, cytotoxic pharmaceuticals, and radioactive isotopes. This complexity is compounded by a rigorous, multi-layered regulatory framework that intersects European Union directives with German federal laws (Bund) and state-specific regulations (Länder). The development of a Waste Management Knowledge Graph (WMKG) represents a transformative solution to this challenge. By encoding regulatory logic and scientific data into a semantic network, institutions can move from manual, error-prone compliance processes to automated, intelligent systems that ensure safety and optimize resource recovery.

The necessity for such a system arises from the inherent ambiguity in waste classification. A single vial containing a chemical residue might be classified as hazardous chemical waste, cytotoxic waste, or standard glass waste depending on the specific active ingredient, its concentration, and the presence of biological contaminants. Traditional "flat" databases or PDF-based guidelines fail to capture the conditional logic required to make these distinctions dynamically. A knowledge graph, however, can link the physical object (the vial) to its chemical properties (via PubChem), its biological risk (via DSMZ/TRBA), and its legal disposal path (via LAGA M18).

This report provides an exhaustive architectural blueprint for constructing a WMKG tailored to the German academic context. It analyzes the necessary data sources, ranging from the LAGA M18 enforcement aids to the PubChem PUG REST API, and defines the ontological structures required to harmonize these disparate datasets. The goal is to provide data engineers, environmental safety officers, and compliance architects with the granular information needed to build a system that not only ensures compliance with the *Abfallverzeichnis-Verordnung* (AVV) but also integrates seamlessly with laboratory inventory systems to protect the health of researchers and the integrity of the environment.

2. Regulatory Backbone: The LAGA M18 and AVV

Framework

The foundational layer of any waste management knowledge graph in Germany is the regulatory framework that defines what constitutes waste and how it must be classified. The central document in this domain is the **LAGA Mitteilung 18** (LAGA M18), titled "Vollzugshilfe zur Entsorgung von Abfällen aus Einrichtungen des Gesundheitsdienstes" (Enforcement Aid for the Disposal of Waste from Healthcare Facilities).[1] This document serves as the authoritative interpretation of the European Waste Catalogue (EWC), implemented in German law as the **Abfallverzeichnis-Verordnung** (AVV).[3]

2.1 The Semantic Hierarchy of the AVV

The AVV codes function as the primary identifiers, or "Foreign Keys," in the knowledge graph, linking the physical reality of waste generation to the legal obligations of disposal. The graph must model Chapter 18 of the AVV, which covers "Wastes from Human or Animal Health Care and/or Related Research."

The classification logic within LAGA M18 is nuanced and relies heavily on the concept of "protection targets" (Schutzziele)—specifically, infection prevention and occupational safety. The knowledge graph must capture these semantic distinctions to automate classification.

2.1.1 AVV 18 01 01: The Logic of Sharps

The code AVV 18 01 01 covers "sharps" (spitze oder scharfe Gegenstände), such as cannulas, scalpels, and broken glass. In a semantic model, this category represents a physical hazard rather than a purely biological one.
The LAGA M18 stipulates that the primary risk here is injury, which could lead to infection, rather than the infection risk inherent in the waste itself.2 Therefore, the graph logic for assigning 18 01 01 must prioritize the physical form of the object.

- **Ontological Rule:** IF Object.hasForm(Sharp) OR Object.hasForm(Pointed) -> Assign WasteCode 18 01 01.
- **Containment Constraints:** The classification triggers a strict containment requirement. The graph must link this node to **ISO 23907** compliant containers.[4] These containers are characterized by their resistance to penetration and their ability to be permanently closed. The data from manufacturers indicates that these containers must meet specific impact and puncture tests.[6]
- **Disposal Pathway:** While often incinerated, the critical attribute is the "irreversible closure" before transport. The graph must encode a procedural rule that prevents the reopening of these containers once sealed, aligning with TRBA 250 safety guidelines.[8]

2.1.2 AVV 18 01 02: Anatomical and Ethical Considerations

This category includes body parts, organs, and blood bags.[2]

- **Semantic Attribute:** The graph must include an attribute for EthicalRequirement. The

disposal of body parts is governed not just by hygiene laws but by ethical standards (Ethik) requiring respectful destruction, typically through incineration (Verbrennung).

- **Separation Logic:** LAGA M18 mandates that these wastes are collected separately at the point of generation. This implies a "Location Constraint" in the graph: Waste classified as 18 01 02 cannot share a collection vessel with 18 01 04 or 18 01 03* at the source.
- **Fluid Constraints:** Blood bags (Blutbeutel) fall here, but only if they are filled. Empty bags might be reclassified. The graph needs a FillStatus property to differentiate between a full blood bag (18 01 02) and an empty one, which might be 18 01 04 depending on residual contamination.

2.1.3 AVV 18 01 03*: The High-Risk Infectious Stream

The asterisk (*) denotes hazardous waste. This is the most strictly regulated stream, defined as waste containing viable micro-organisms or their toxins which are known or reliably believed to cause disease in humans or other living organisms.[9]

- **Dynamic Classification:** Unlike chemicals, which have static hazard properties, the "infectious" nature of waste is context-dependent. It depends on the pathogen's Risk Group (see Section 3). The KG must allow for dynamic inference: IF Pathogen.RiskGroup >= 3 THEN WasteCode = 18 01 03*.
- **Transport Linkage:** This code is strongly coupled with transport regulations. Waste classified as 18 01 03* almost always triggers **UN 3291** (Clinical Waste, unspecified, n.o.s.) or, in extreme cases (Category A substances), **UN 2814**.[11] The graph must generate the appropriate ADR transport labels (Class 6.2).
- **Treatment Requirement:** The graph must enforce a "Pre-treatment" rule. Many universities require autoclaving of such waste before it leaves the facility, effectively "deactivating" it to a lower risk category (though the code might remain for traceability).

2.1.4 AVV 18 01 04: The "B-Müll" Distinction

This code covers non-hazardous medical waste (dressings, casts, linen). The distinction between 03* and 04 is a frequent source of confusion and cost inefficiency.

- **The "Specific Requirements" Test:** LAGA M18 clarifies that waste is 18 01 04 if *no specific requirements* for infection prevention are needed outside the medical area.[2] This means waste contaminated with standard pathogens (like *Staphylococcus aureus* in a wound dressing) is generally 18 01 04, not 03*, unless the quantity of blood/fluid suggests a higher risk of spread.
- **Economic Implication:** 18 01 04 is significantly cheaper to dispose of (often waste-to-energy). The KG acts as a cost-optimization tool by preventing the "over-classification" of safe waste into the expensive 03* stream.
- **Data Integration Strategy:** The KG should query a "Pathogen Allow-list" derived from hygiene plans. If a pathogen is on the list (e.g., standard skin flora), the waste defaults to 18 01 04.

2.2 Structured Data Sources for AVV

To populate the WasteCode class, we cannot rely on manual entry. We require authoritative datasets.

- **Source:** The Federal Statistical Office (Destatis) and state environmental portals (e.g., Schleswig-Holstein Open Data) provide the AVV catalog in machine-readable formats (XML/CSV).[13]
- **ETL Process:**
  1. Ingest the full AVV dataset.
  2. Filter for Chapter 18.
  3. Enrich the nodes with textual descriptions from the LAGA M18 PDF [2] using NLP to extract keywords like "blood," "secretions," "cytostatic," and "sharp."
  4. This creates a searchable index where a user typing "blood bag" is automatically guided to 18 01 02.

3. Biological Agents: The BioStoffV and TRBA Data Backbone

For university laboratories, particularly in microbiology and virology, waste classification is driven by the **Biostoffverordnung (BioStoffV)**. This ordinance mandates that all biological agents be classified into Risk Groups (1-4). The Knowledge Graph must serve as a biological oracle, instantly retrieving the risk group of any organism to determine the waste classification.

3.1 The Technical Rules for Biological Agents (TRBA)

The TRBA documents are the primary data source for this domain. They provide the official German classification lists.[8]

3.1.1 TRBA 466: The Prokaryotic Taxonomy

TRBA 466 governs the classification of Bacteria and Archaea.[17]

- **Data Structure:** The data is presented in tabular format containing Genus, Species, and Risk Group (RG).
- **Conditional Logic:** The graph must handle complexity beyond simple key-value pairs.
  - *Pathogenicity Markers:* Many organisms are listed with notations like "Z" (Zoonosis) or "T" (Toxin). For example, *Clostridium botulinum* is RG 2 but produces a highly potent toxin. Waste containing this organism might trigger both biological (18 01 03*) and chemical (18 01 06*) waste rules depending on the toxin concentration.[8]
  - *Strain Specificity:* The KG must distinguish between strains. *Escherichia coli* K12 is RG 1 (safe), while *E. coli* O157:H7 is RG 3** (hazardous). The data model must support Strain as a subclass of Organism.

- **Ingestion Strategy:** Since BAuA provides these as PDFs, a PDF-to-JSON parsing pipeline is necessary. The parser must recognize the columns "Spezies", "Risikogruppe", and "Bemerkungen" to populate the graph.[18]

3.1.2 TRBA 460: Fungi

This rule classifies fungi.[15]

- **Disposal Implication:** Fungi in RG 2 or higher (e.g., *Aspergillus fumigatus*) generally require thermal inactivation (autoclaving) before disposal. The KG must link RG 2+ fungal nodes to an Instruction: Autoclave property.
- **Allergenic Potential:** TRBA 460 also notes allergenic potential ("A"). While this doesn't change the waste code, it triggers "Respiratory Protection" requirements for the waste handler, which the KG should output as safety guidance.

3.1.3 TRBA 462: Viruses

This rule classifies viruses and TSE agents.[16]

- **The "Star" Notation:** TRBA 462 uses a nuanced notation: 3** and 3***. This indicates that while the virus is Risk Group 3, the risk of airborne transmission is limited (3**) or non-existent (3***).
- **Graph Logic:** This is crucial for laboratory infrastructure but also for waste. An 18 01 03* waste stream for a 3** virus might not require the same hermetically sealed transport logistics as a full RG 4 virus (e.g., Ebola). The graph allows the university to tailor safety measures to the specific viral phenotype without violating the law.

3.2 Digital Data Sources: DSMZ and ZKBS

To avoid the fragility of PDF scraping, the KG should integrate robust digital databases.

3.2.1 DSMZ (German Collection of Microorganisms)

The **DSMZ** is one of the world's largest biological resource centers.

- **Data Richness:** Their catalogue contains detailed biosafety data for over 80,000 strains.[22]
- **API/Access:** While a public open API is not explicitly advertised as "free for all," the search functionality allows for structured queries. The "Safety Information" provided for each strain acts as a ground-truth verification for the KG. If a lab claims a strain is RG 1, but DSMZ lists it as RG 2, the KG can flag a compliance error.
- **Delivery Restrictions:** DSMZ data regarding shipping restrictions (e.g., prohibition of UN 3373 to certain countries) [23] informs the "Transport" module of the KG.

3.2.2 ZKBS (Central Committee on Biological Safety)

The **ZKBS** database is specific to genetic engineering (Gentechnik).[24]

- **Divergence from TRBA:** Sometimes, the ZKBS classification (for GMO work) differs from the TRBA classification (for occupational safety). The KG must store both values.
- **Excel Export:** The ZKBS offers an "Organismenliste" as an exportable file (often Excel/CSV).[25] This is a high-value dataset for bulk ingestion into the KG.
- **Gene Transfer:** The KG can model "Donor" and "Recipient" organisms. If a safe recipient (RG 1) receives a virulence gene from a donor (RG 3), the resulting GMO waste might be RG 2 or 3. The ZKBS database provides the rules for these combinations.

3.3 Semantic Logic for Biological Waste Inference

The Knowledge Graph utilizes this data to perform "Risk Inference."

1. **Input:** User enters "Culture of *Salmonella typhi*".
2. **Lookup:** KG queries TRBA 466 node for *Salmonella typhi*.
3. **Result:** Risk Group 3**.
4. **Rule Execution:**
    - RiskGroup >= 3 -> Waste Code **18 01 03***.
    - Transmission!= Airborne -> Container **Yellow UN 3291 Bin**.
    - Requirement -> **Autoclave on site** before release to third-party disposal.

4. Chemical Waste: GHS Integration and PubChem API

Chemical waste in university laboratories is highly heterogeneous. It includes organic solvents, inorganic acids, heavy metals, and complex reaction mixtures. The primary challenge is segregation: mixing incompatible chemicals can cause explosions or toxic gas release. The KG automates this sorting using **GHS (Globally Harmonized System)** data.

4.1 PubChem PUG REST API: The Chemical Oracle

For chemical data, **PubChem** is the most reliable and accessible source.[26]

4.1.1 API Integration Strategy

The KG interacts with PubChem via the **PUG REST** and **PUG View** APIs.

- **Endpoint:** https://pubchem.ncbi.nlm.nih.gov/rest/pug_view/data/compound/{CID}/JSON?heading=GHS%20Classification.[29]
- **Workflow:**
    1. User inputs "Methanol" or CAS "67-56-1".
    2. KG resolves this to PubChem CID 887.
    3. KG fetches the GHS JSON object.
    4. KG parses the **Hazard Codes** (H-Codes) and **Precautionary Codes** (P-Codes).

4.1.2 Data Extraction and Logic Mapping

The extracted H-codes drive the sorting logic (Abfall-ABC).[30]

H-Code	Meaning	University Waste Stream (KG Class)	Container
H224, H225	Extremely/Highly Flammable Liquid	HalogenFreeSolvent	10L PE Conductive Canister
H226	Flammable Liquid	HalogenFreeSolvent	10L PE Conductive Canister
H350	May cause cancer	CarcinogenicWaste	Sealed Hazardous Waste Drum
H290	May be corrosive to metals	InorganicAcid or Base	Glass/PE Bottle (Check pH)
H400, H410	Toxic to aquatic life	HeavyMetalWaste (if metal)	Separate Collection

- **The Halogen Rule:** A critical sorting rule in Germany is the separation of halogenated vs. non-halogenated solvents.[33] The KG checks the molecular formula (fetched from PubChem) for Cl, Br, F, or I.
    - IF Formula contains (Cl OR Br OR F) AND IsOrganic -> WasteStream = HalogenatedSolvent.
    - *Reasoning:* Halogenated solvents require expensive high-temperature incineration (>1100°C) to prevent dioxin formation. Non-halogenated solvents can be burned for energy recovery at lower cost. The KG enforces this separation to optimize costs.

4.2 GESTIS and REACH: European Compliance

While PubChem is global, **GESTIS** and **ECHA** provide the specific EU/German regulatory context.

4.2.1 GESTIS Stoffdatenbank

Maintained by the German Social Accident Insurance (IFA), GESTIS reflects the **Gefahrstoffverordnung (GefStoffV)**.[34]

- **Role in KG:** Used for validation. If PubChem lists a chemical as "Warning" but GESTIS lists it as "Danger" under German law, the KG prioritizes the stricter GESTIS classification.
- **ZVG Number:** The KG should store the ZVG number (GESTIS identifier) to link directly to

German safety data sheets.

4.2.2 ECHA (REACH)

The European Chemicals Agency (ECHA) manages REACH data.[35]

- **SVHC Identification:** The KG checks if a chemical is a "Substance of Very High Concern" (SVHC). Disposal of SVHCs often requires specific reporting.
- **Authorization:** ECHA data confirms if the lab is even allowed to use the substance.

4.3 Chemical Compatibility Matrix

The KG effectively prevents accidents by cross-referencing H-codes.

- **Inference Rule:** IF WasteA has H271 (Oxidizer) AND WasteB has H225 (Flammable) -> INCOMPATIBLE.
- **User Alert:** If a user tries to add "Nitric Acid" to a "Solvent" canister in the system, the KG triggers a "STOP" alert with a red GHS incompatibility warning.

5. Pharmaceutical Waste: Cytostatics and CMR Substances

Pharmaceutical waste is a high-stakes domain due to the presence of **Cytotoxic and Cytostatic drugs** (CMR drugs), which pose severe health risks to waste handlers and environmental risks if leached into groundwater.

5.1 The "Blue Stream" (AVV 18 01 08*)

The AVV distinguishes between:

- *18 01 08:** Cytotoxic and cytostatic medicines (Hazardous).
- **18 01 09:** Medicines other than those mentioned in 18 01 08 (Non-Hazardous).

5.1.1 Defining "Cytotoxic" via TRGS 525

The Technical Rule **TRGS 525** defines which drugs fall into 18 01 08* based on their CMR properties (Carcinogenic, Mutagenic, Reprotoxic).[36]

- **Graph Logic:** The KG identifies a drug as 18 01 08* if:
  - It is listed in **NIOSH Table 1** (Antineoplastic).
  - It carries H-codes **H340, H350, H360** (CMR Category 1A/1B).
  - It is not merely a "potential" hazard (like some hormones in NIOSH Table 2) but a confirmed hazard under German BGW guidelines.[38]

5.2 Data Integration: NIOSH and BGW Lists

The KG ingests two primary lists to make this decision.

1. **NIOSH List:** The US standard.[39] It categorizes drugs into:
    - *Group 1:* Antineoplastics (e.g., Cisplatin, Doxorubicin). -> **Always 18 01 08***.
    - *Group 2:* Non-antineoplastic hazardous drugs (e.g., Carbamazepine). -> **Context dependent**.
    - *Group 3:* Reproductive hazards (e.g., Warfarin). -> **Context dependent**.
2. **BGW List (German):** The Berufsgenossenschaft list (e.g., BGW 09-19-042) is the local authority.[36] It is often more conservative/specific than NIOSH regarding what requires high-temperature incineration.

5.3 Disposal Logic: The "Anhaftung" Rule

A crucial economic nuance in German law is the "Anhaftung" (adhesion) rule for 18 01 04.[9]

- **The Rule:** Items *trace-contaminated* with cytostatics (gloves, swabs, empty infusion sets) can often be disposed of as 18 01 04 (Blue/Black bin, non-hazardous), provided no significant liquid or solid residue remains.
- **Graph Decision Tree:**
    - Is Item a Drug Vial? -> Yes -> Is it Empty? -> No -> **18 01 08***.
    - Is Item PPE (Glove)? -> Yes -> Is it Dripping/Saturated? -> No -> **18 01 04**.
    - Is Item Bulk Drug? -> Yes -> **18 01 08***.
- **Containerization:** 18 01 08* requires rigid, sealed, blue containers (often called "Cytoboxes" or "C-Fässer") to prevent aerosolization during handling.[42]

6. Radioactive Waste: Physics and Decay Logic

For nuclear medicine and physics departments, waste is governed by the **Strahlenschutzverordnung (StrlSchV)**. This stream is unique because waste can "become" non-waste through the passage of time (Decay).

6.1 The Concept of Clearance (Freigabe)

Radioactive waste is not always radioactive waste. If its activity concentration (Bq/g) falls below the "Freigabewerte" in **Anlage 4 StrlSchV**, it can be cleared and disposed of as standard waste.[43]

6.2 Nuidle Data Tables

The KG must incorporate a digitized version of Anlage 4.

- **Table Structure:**
    - Nuclide (e.g., P-32, I-125, H-3).
    - Half-life ($T_{1/2}$).
    - Clearance Level (Freigrenze) for solid waste (Bq/g).
    - Clearance Level for liquids (Bq/g).

Selected Isotope Data for KG Implementation [43]:

Isotope	Half-Life (T1/2)	Clearance Level (Solid)	Clearance Level (Liquid)
H-3	12.3 years	1,000 Bq/g	1,000 Bq/g
C-14	5,730 years	1,000 Bq/g	100 Bq/g
P-32	14.3 days	1,000 Bq/g	100 Bq/g
I-125	59.4 days	100 Bq/g	10 Bq/g
Tc-99m	6.0 hours	100 Bq/g	10 Bq/g

6.3 The Decay Calculator Module

The KG utilizes a mathematical module to optimize storage duration (Abklingzeiten).

- Formula: The time t required to reach the clearance level L from an initial activity A_0 is:

  $$t = \frac{\ln(A_0 / L) \cdot T_{1/2}}{\ln(2)}$$
- **Use Case:** A lab generates P-32 waste with an activity of 1 MBq/g. The limit is 1 kBq/g.
  - Ratio $A_0/L = 1000$.
  - $\ln(1000) \approx 6.9$.
  - $t \approx 6.9 \times 14.3 / 0.693 \approx 142$ days.
- **System Action:** The KG schedules a notification for 142 days in the future: "Container P32-001 ready for clearance measurement." This prevents premature (illegal) disposal and minimizes storage costs for short-lived isotopes.

7. Transport and Packaging: The ADR Connection

Waste does not teleport; it must be transported. This falls under the **ADR** (European Agreement concerning the International Carriage of Dangerous Goods by Road). The KG must map AVV codes to UN Numbers.

7.1 Key UN Numbers and Packaging

The KG links waste types to specific UN packaging requirements.[11]

UN 3291: Clinical Waste

- **Mapping:** Corresponds to AVV 18 01 03*.
- **Packaging:** Requires **Type 4H2** (Solid Plastic Box) or **Type 1H2** (Plastic Drum).
- **Testing:** Packaging must pass drop tests (1.2m) and leakproofness tests.
- **Labeling:** "Biohazard" symbol + "UN 3291".

UN 2814 / UN 2900: Category A Infectious Substances

- **Mapping:** High-risk pathogens (e.g., Ebola culture).
- **Packaging:** Triple packaging system (Primary leakproof, Secondary leakproof, Rigid outer). Strict P620 packing instruction.

UN 3373: Biological Substance, Category B

- **Mapping:** Diagnostic specimens (samples), not usually bulk waste, but relevant for inter-lab transport.
- **Packaging:** P650 instruction.

7.2 Sharps Containers: ISO 23907

For AVV 18 01 01 (Sharps), the KG enforces **ISO 23907** compliance.[4]

- **Technical Spec:**
  - **Penetration Resistance:** Must withstand >15N force (depending on wall thickness).
  - **Closure:** Must have a temporary (day-use) and permanent (final) closure.
  - **Color:** Typically yellow with a red lid (signaling hazard).
- **Graph Validation:** If a user selects "Sharps" waste, the system validates that the selected container SKU corresponds to an ISO 23907 certified product.

8. Knowledge Graph Architecture and Implementation

8.1 Ontology Design

The ontology harmonizes the domains of waste, chemistry, and biology. It should reuse existing standards where possible and extend them for local rules.

8.1.1 Reused Ontologies

- **EnvO (Environment Ontology):** Provides classes for waste disposal process, medical waste, and hazardous waste.[47] It connects the university waste stream to broader ecological concepts.
- **EWC-Onto:** Models the hierarchical structure of the European Waste Catalogue (Chapter -> Subchapter -> Code).[49]
- **ChEBI:** For chemical entities.

8.1.2 Custom University Waste Ontology (UWO)

We define a custom schema to bridge gaps.

- **Class:** uwo:WasteStream
  - uwo:hasAVVCode: Link to EWC-Onto.
  - uwo:requiresContainer: Link to uwo:Container.
  - uwo:hasRiskLevel: "High", "Medium", "Low".
- **Class:** uwo:Container
  - uwo:certification: "UN 3291", "ISO 23907".
  - uwo:material: "PP", "HDPE".
- **Class:** uwo:RegulatoryRule
  - uwo:source: "LAGA M18", "TRGS 525".
  - uwo:logic: SHACL constraints or SWRL rules.

8.2 Data Ingestion Pipeline (ETL)

Building the graph requires a robust pipeline.

1. **Static Data (The "Cold" Layer):**
   - *Source:* AVV XML, TRBA PDFs.
   - *Process:* Python scripts using PyPDF2 or Tabula to scrape tables from TRBA 466/462. Map species names to NCBI Taxonomy IDs for normalization.
   - *Update Frequency:* Quarterly (when regulations update).
2. **Dynamic Data (The "Hot" Layer):**
   - *Source:* PubChem API, NIOSH Lists.
   - *Process:* On-demand querying. When a chemical is added to inventory, the system queries PubChem PUG REST to fetch H-codes. This data is cached in the graph.
   - *Update Frequency:* Real-time or Daily.

8.3 Example RDF Triple Construction

Code snippet

```
@prefix uwo: <http://university-waste.de/ontology#>.
@prefix avv: <http://w3id.org/ewc-onto/>.
@prefix pubchem: <http://rdf.ncbi.nlm.nih.gov/pubchem/compound/>.

# Instance: A specific waste bag from the Virology Lab
uwo:Waste_ID_12345 a uwo:InfectiousWaste ;
   uwo:origin "Virology_Lab_Building_C" ;
   uwo:containsOrganism "Influenza A virus (H1N1)" ;
```

```
    uwo:hasRiskGroup "2" ;
    uwo:classificationStatus "3**" ; # TRBA 462 nuance
    avv:hasCode "18 01 03*" ;
    uwo:requiresTransport "UN 3291" ;
    uwo:disposalInstruction "Autoclave_Before_Release".

# Instance: A container of Acetone
uwo:Waste_ID_67890 a uwo:ChemicalWaste ;
    uwo:containsChemical pubchem:CID180 ; # Acetone
    uwo:hasHCode "H225" ;
    uwo:isHalogenated false ;
    avv:hasCode "14 06 03*" ; # Other solvents and solvent mixtures
    uwo:requiresContainer uwo:Canister_Conductive_10L.
```

9. Conclusion

The construction of a Waste Management Knowledge Graph for German universities is a sophisticated engineering task that demands the synthesis of law, science, and logistics. By rigorously mapping the **LAGA M18** directives to the physical reality of lab waste, and enriching this model with live data from **PubChem**, **TRBA**, and **StrlSchV**, institutions can achieve a new standard of safety and efficiency.

The architecture proposed here moves beyond simple compliance; it creates a "semantic safety net." It catches the dangerous mixture of chemicals before they are poured; it flags the infectious pathogen before it leaves the lab; and it times the disposal of radioactive isotopes to the day. In doing so, it protects the institution from legal liability and, most importantly, protects the people and the environment that sustain the academic mission. The path forward involves the systematic ingestion of the datasets identified—NIOSH lists, TRBA tables, PubChem properties—and their crystallization into the ontological structures defined in this report.

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